2026 International Conference on Robotic Welding, Intelligence and Automation (RWIA 2026) & The 16th Chinese Conference on Robotic Welding (CCRW 2026)

December 11 – 13, 2026, Shanghai, China

Oct. 20, 2026

Call for Papers

Robotic welding is moving beyond fixed teach-and-playback workflows toward systems that combine sensing, planning, and feedback control. In production, variation in joint geometry and fit-up, changes in the work environment, uncertainty in process conditions, and thermal distortion continue to challenge reliable automation. Advances in multimodal sensing, physics-informed and data-driven modeling, foundation models, embodied AI, and digital twins offer opportunities to improve adaptation. Research is needed to establish when these methods are robust and effective in practical welding tasks.

RWIA 2026 and CCRW 2026 will be held in Shanghai on December 11–13, 2026. The RWIA series began in Shanghai in 2002, and the 2026 event is planned as its ninth meeting. The joint conference will bring together researchers and practitioners working on robotic welding, intelligent robots, and advanced manufacturing. Invited speakers will discuss recent research and technical advances, industry participants will present welding robot technologies, and attendees will exchange experience from research and industrial applications.

Selected English-language manuscripts that pass peer review are planned for publication in 2027 in Transactions on Intelligent Welding Manufacturing (TIWM), a Springer Nature book series. The series is indexed in Scopus, and published volumes are available on Springer Nature Link. Publication dates and volume allocation are subject to editorial decisions.

Topics include, but are not limited to:

S01: Intelligent robotic welding and manufacturing systems
S02: Special-purpose welding robots and embodied AI
S03: Autonomous planning and decision-making for robotic welding
S04: Multimodal sensing and process-state estimation in welding
S05: Weld-quality monitoring, assessment, and feedback control
S06: Foundation models and knowledge-driven methods for welding
S07: Digital twins, process modeling, simulation, and optimization
S08: Multi-robot coordination and welding in challenging environments
S09: Intelligent, digital, and flexible welding equipment
S10: Related topics in intelligent manufacturing

Paper Submission:

Manuscripts must be written in English. An English manuscript based on work submitted to CCRW 2026 may be considered only if it complies with TIWM's originality, prior-publication, overlap, and copyright requirements. Authors should disclose any related submission or prior publication to the editors and contact the conference secretariat if they need guidance on translation. Prepare a single Word manuscript using the TIWM SPLNPROC template and follow the series' formatting instructions. Submit the manuscript through the TIWM online system and email a copy to the conference secretariat. The recommended maximum lengths are 20 pages for Feature Articles, 12 pages for Regular Papers, and 6 pages for Technical Notes.

Online submission: https://ocs.springer.com/ocs/en/home/TIWM2017

Conference secretariat: rwlab21@sjtu.edu.cn

Each planned volume is expected to contain approximately 15 papers. Manuscripts may be submitted between the full-paper deadlines below; acceptance and placement in a volume depend on peer review and the editorial schedule.

Important Dates:

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RWIA 2026 abstract submission:	Nov. 30, 2026
First full-paper submission deadline	Jan. 30, 2027
Second full-paper submission deadline:	Apr. 30, 2027

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Program Committee of RWIA 2026 and CCRW 2026

Robotics & Automation Committee of CWS

The Organizations of RWIA '2026/CCRW '2026 is omitted here.